MODULE 5.1

Secure Networks with Firewalls

Firewall Types | Packet Filtering & Stateful Inspection

Presented by: Kudzaishe Majeza

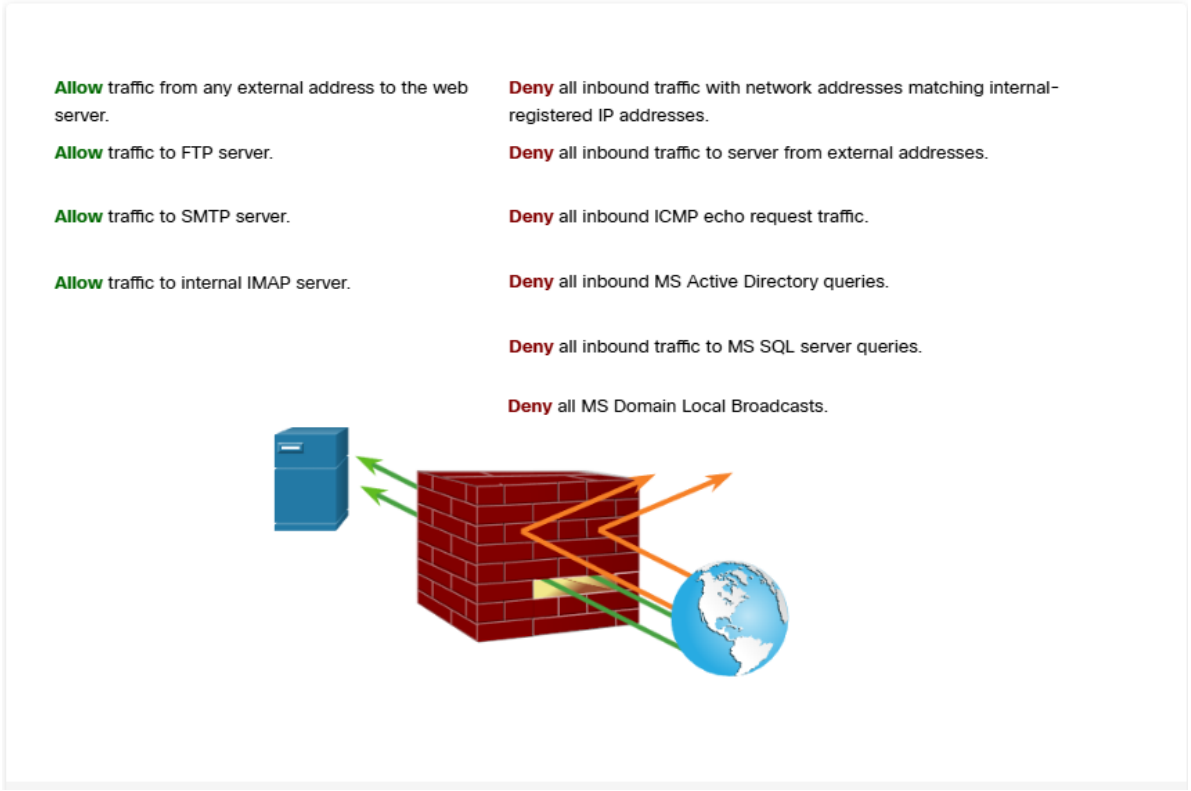
Cisco Networking Academy | 2026

Topics:

- 5.1.1 Firewalls
- 5.1.2 Types of Firewalls
- 5.1.3 Check Your Understanding — Identify the Type of Firewall
- 5.1.4 Packet Filtering Firewall Benefits and Limitations
- 5.1.5 Stateful Firewall Benefits and Limitations
- Summary
- References

Firewalls

- **What it is:** a hardware or software system that enforces an access control policy between two networks — everything crossing the boundary is inspected against that policy.
- **Network-based vs. host-based:** a network firewall is a dedicated appliance (or router ACL) protecting a whole segment; a host-based firewall is software running on one device, like Windows Defender Firewall, protecting just that machine.
- **Shared properties:** resistant to attacks itself, sits as the only transit point between the two networks so nothing can sneak around it, and enforces the access control policy consistently.
- **Why it matters:** it hides sensitive hosts and services from the outside, rejects malformed or suspicious packets, and centralizes security management instead of securing every host individually.



Abbreviation: ACL = Access Control List.

Types of Firewalls

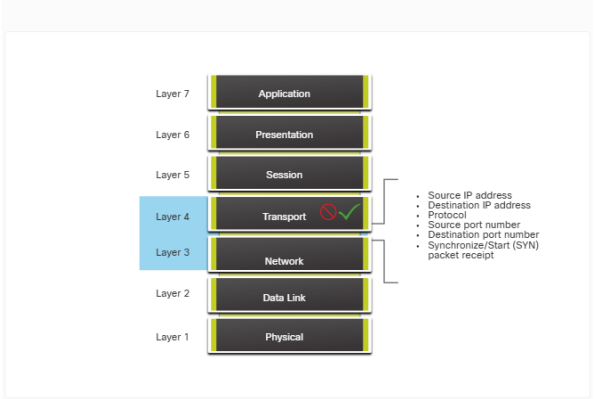
- **Packet filtering:** the simplest type — checks each packet's source/destination address and port against a static rule list, layer 3–4 only, no memory of past packets.

- **Stateful:** also layer 3–4, but tracks every active connection in a state table, so it can tell a genuine reply apart from an unsolicited new connection.

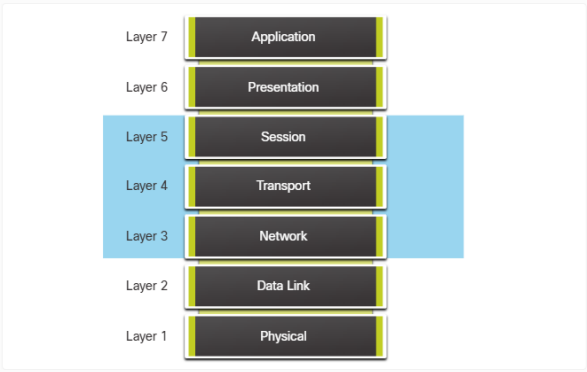
- **Application / proxy:** operates up at layer 7, actually reading application data — it can tell a real HTTP request from traffic just disguised as port 80.

- **Next-Generation (NGFW):** combines stateful inspection with deep packet inspection, intrusion prevention, and application awareness in one device.

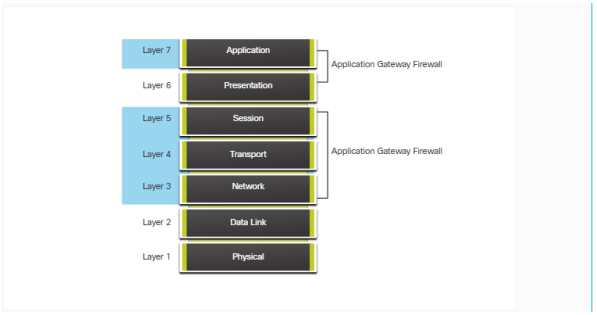
Packet Filtering



Stateful



Proxy / Application



Next-Gen (NGFW)



Check Your Understanding

Read the behaviour, then name the firewall type responsible for it:

Q: Blocks a packet purely because its destination port is 23 (Telnet), with no regard for whether a session is already open.

A: Packet Filtering Firewall

Q: Lets a reply packet back in only because it matches an entry already sitting in the connection table, and drops it otherwise.

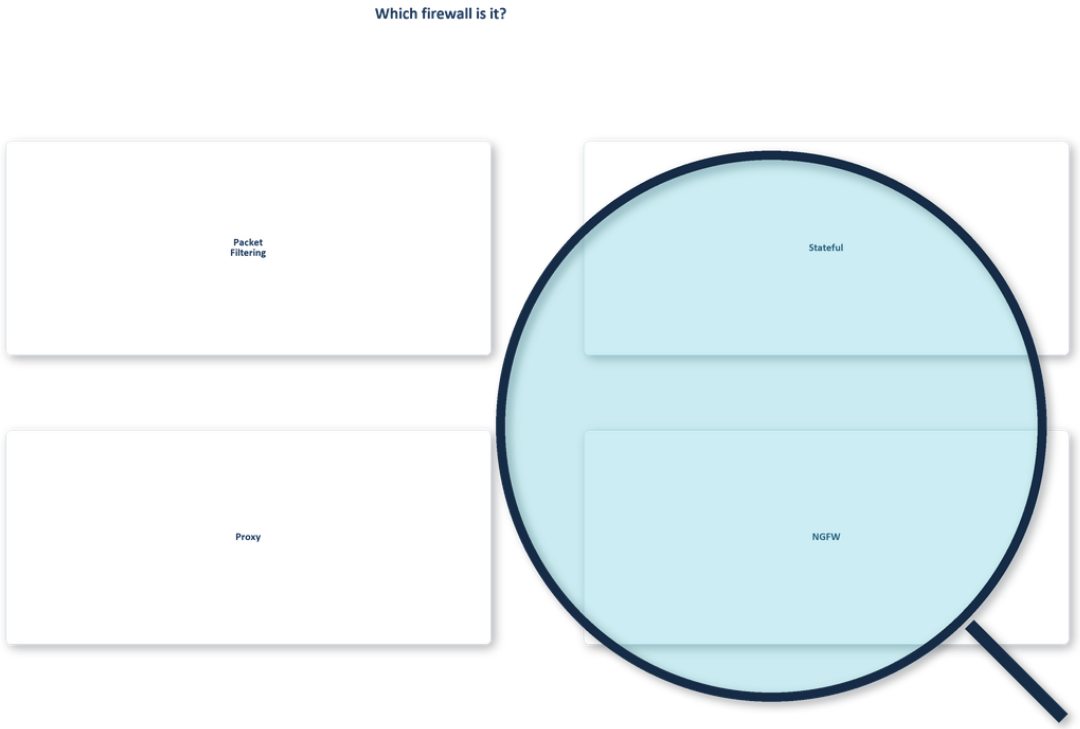
A: Stateful Firewall

Q: Reads the actual HTTP request content to catch traffic that's disguised as web traffic but isn't really HTTP at all.

A: Application / Proxy Firewall

Q: Combines connection-state tracking with intrusion prevention and app-awareness in a single appliance.

A: Next-Generation Firewall (NGFW)



Packet Filtering: Benefits and Limitations

Benefits

– Built into practically every router — no extra hardware or cost.

– Very low performance impact, since it only reads packet headers.

– Simple to reason about: it's just a list of source / destination / port rules — exactly how a standard or extended ACL works.

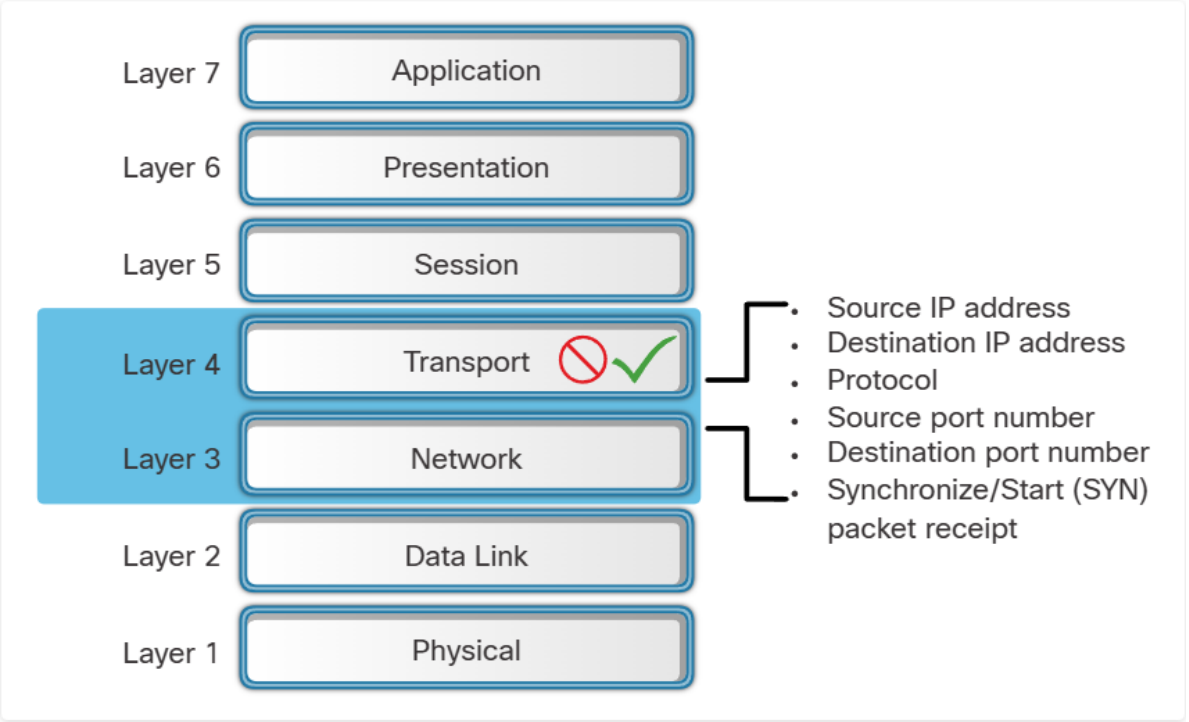
Limitations

– Stateless — it has no memory of earlier packets, so it can't tell a genuine reply from a forged first-contact packet.

– Can't see inside the payload, so it can't filter based on application content.

– Vulnerable to IP spoofing, since source address is trusted at face value.

– Any new application support means manually adding more static rules.



Abbreviations: IP = Internet Protocol. ACL = Access Control List.

Stateful Firewall: Benefits and Limitations

Benefits

– The primary filtering method in production networks today.

– Matches traffic against the full context of a connection, not one isolated packet.

– Produces richer logs, since each entry can reference a whole session.

– Effective against spoofing and session-hijacking a stateless ACL would miss.

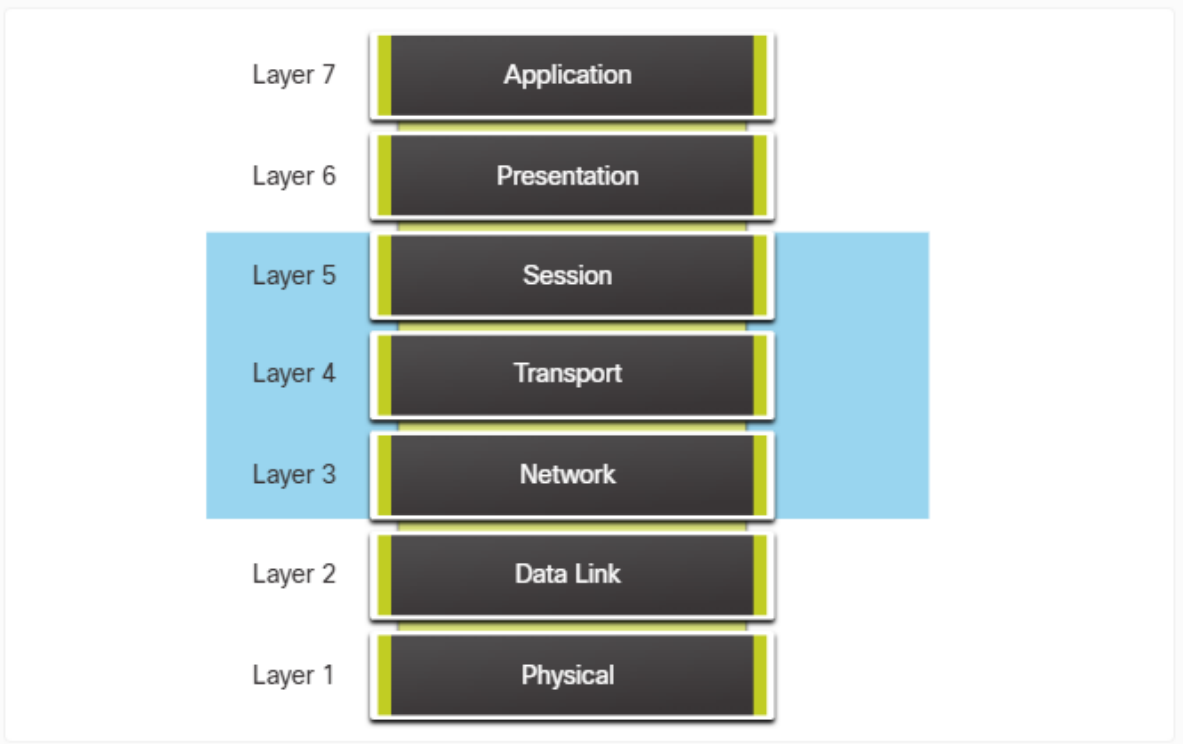
Limitations

– Still can't inspect application-layer payload, so it can't stop every application-layer attack.

– Not every protocol is connection-oriented — UDP and ICMP have no real "session" to track.

– More resource-intensive — maintaining a state table costs memory and CPU.

– A flood of half-open TCP connections can exhaust the state table — its own DoS exposure.



Abbreviations: UDP = User Datagram Protocol. ICMP = Internet Control Message Protocol. DoS = Denial of Service.

Summary

- A firewall enforces an access control policy at the one point all traffic between two networks must cross.
- **5.1.1 —** firewalls can be network-based (dedicated appliance) or host-based (software on one device), but both resist attack and enforce policy consistently.
- **5.1.2 —** packet filtering and stateful firewalls work at layers 3–4; application/proxy firewalls work at layer 7; NGFWs combine all of it in one device.
- **5.1.4 —** packet filtering is cheap and fast but stateless, so it can't verify a session or see inside the payload.
- **5.1.5 —** stateful inspection is the modern default — session-aware and harder to spoof, at the cost of more resources and its own DoS exposure via the state table.

References

Cisco Networking Academy — Network Defense, Module 5.1: Secure Networks with Firewalls

https://www.netacad.com

Cisco — Firewalls (Security Technologies Overview)

https://www.cisco.com/c/en/us/products/security/firewalls/what-is-a-firewall.html

NIST — Guidelines on Firewalls and Firewall Policy (SP 800-41)

https://csrc.nist.gov/publications/detail/sp/800-41/rev-1/final